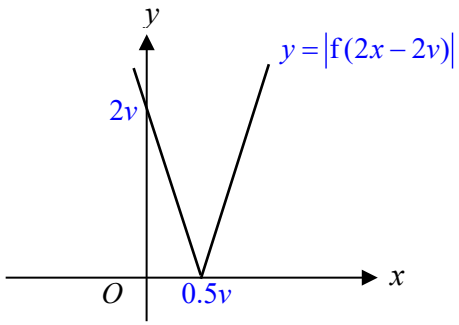
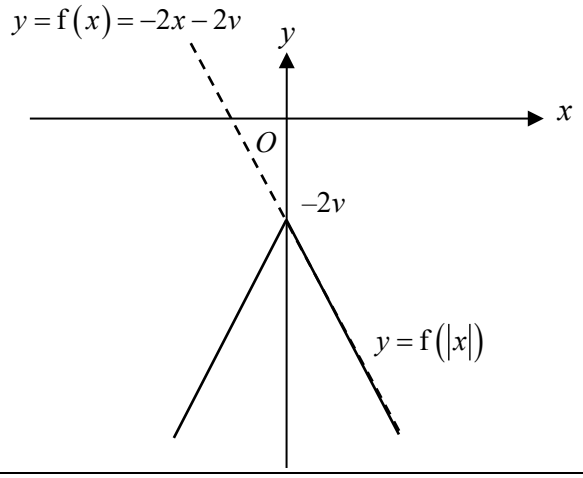
VJC 2026 H2 Math Prelim Paper 2 (Solution)

1	Solution
(a)	**Method 1:** $y = \lvert ax + b \rvert$ Subt $(0, 2v)$: $2v = \lvert a(0) + b \rvert \quad \Rightarrow \quad \lvert b \rvert = 2v$ $\Rightarrow \quad b = \pm 2v$ Subt $(-v, 0)$: $0 = \lvert a(-v) + b \rvert \quad \Rightarrow \quad -av + b = 0$ $\Rightarrow \quad a = \frac{b}{v}$ Hence, $a = 2$ and $b = 2v$ or $a = -2$ and $b = -2v$ The expressions are: $\mathrm{f}(x) = 2x + 2v$ and $\mathrm{f}(x) = -2x - 2v$ **Method 2:** $y = \lvert ax + b \rvert = \begin{cases} ax + b \\ -ax - b \end{cases}$ Consider $y = ax + b$: Gradient $a = \frac{0 - 2v}{-v - 0} = 2$, y-intercept $b = 2v$ Hence $y = \lvert ax + b \rvert = \begin{cases} 2x + 2v \\ -2x - 2v \end{cases}$ The expressions are: $\mathrm{f}(x) = 2x + 2v$ and $\mathrm{f}(x) = -2x - 2v$
(b)(i)	$y = \lvert \mathrm{f}(2x - 2v) \rvert$ **Method 1:** 1. Translate $y = \lvert \mathrm{f}(x) \rvert$ by $2v$ units in the positive x-direction 2. Stretch resultant $y = \lvert \mathrm{f}(x - 2v) \rvert$ parallel to the x-axis by factor $\frac{1}{2}$.  **Method 2** Since $a < 0$, $\mathrm{f}(x) = -2x - 2v$ $y = \lvert \mathrm{f}(2x - 2v) \rvert$ $y = \lvert -2(2x - 2v) - 2v \rvert = \lvert -4x + 2v \rvert$

(b) **(ii)**	Since $a < 0,\ \mathrm{f}(x) = -2x - 2v$ 

2	**Solutions**
(a)	If w is a root of $P(z) = 0$, then $w^6 + aw^4 + bw^2 + c = 0$ Consider $P(-w) = (-w)^6 + a(-w)^4 + b(-w)^2 + c = 0$ $= w^6 + aw^4 + bw^2 + c$ $= 0$ Hence $-w$ must also be a root.
(b)	Since all coefficients of the polynomial are real, non-real roots occur in conjugate pairs. Hence, if $1+\mathrm{i}$ is a root, $1-\mathrm{i}$ is also a root. From (a) which states that if w is a root, then $-w$ is also a root. $-(1+\mathrm{i})$ and $-(1-\mathrm{i})$ are also roots. Hence 3 other non-real roots are $1-\mathrm{i}, -1-\mathrm{i}$ and $-1+\mathrm{i}$.
(c)	**<u>Method 1:</u>** $z^6 + az^4 + bz^2 + c$ $= (z-(1+\mathrm{i}))(z-(1-\mathrm{i}))(z-(-1-\mathrm{i}))(z-(-1+\mathrm{i}))(z-k)(z+k)$ Compare constant term: $[-(1+\mathrm{i})][-(1-\mathrm{i})][-(-1-\mathrm{i})][-(-1+\mathrm{i})](-k)(k) = c$ $-4k^2 = c$ $k = \pm\sqrt{-\frac{c}{4}}$ $(\text{where } c < 0)$ Hence, the remaining roots are $\pm\sqrt{-\frac{c}{4}} = \pm\frac{\sqrt{-c}}{2}$.

	Method 2: [expand $(z-w)(z+w)=z^2-w^2$] $z^6+az^4+bz^2+c$ $=(z-(1+\mathrm{i}))(z+(1+\mathrm{i}))(z-(1-\mathrm{i}))(z+(1-\mathrm{i}))(z^2+mz+n)$ $=(z^2-(1+\mathrm{i})^2)(z^2-(1-\mathrm{i})^2)(z^2+mz+n)$ $=(z^2-2\mathrm{i})(z^2+2\mathrm{i})(z^2+mz+n)$ $=(z^4+4)(z^2+mz+n)$ Compare z term: $4m=0 \Rightarrow m=0$ Compare constant: $4n=c \Rightarrow n=\frac{c}{4}$ To get the remaining 2 roots $z^2+\frac{c}{4}=0 \Rightarrow z=\pm\sqrt{-\frac{c}{4}}=\pm\frac{\sqrt{-c}}{2}$

3	**Solution**
(a)	Using ratio theorem, $\overrightarrow{OM}=\frac{\overrightarrow{OA}+\overrightarrow{OC}}{2}=\frac{\underset{\sim}{a}+\underset{\sim}{c}}{2}$ **Method 1:** Given $\underset{\sim}{a}\times\underset{\sim}{b}=\underset{\sim}{b}\times\underset{\sim}{c}$ $\underset{\sim}{a}\times\underset{\sim}{b}-\underset{\sim}{b}\times\underset{\sim}{c}=\underset{\sim}{0}$ $\underset{\sim}{a}\times\underset{\sim}{b}+\underset{\sim}{c}\times\underset{\sim}{b}=\underset{\sim}{0}$ $(\underset{\sim}{a}+\underset{\sim}{c})\times\underset{\sim}{b}=\underset{\sim}{0}$ Since $\underset{\sim}{a}$ and $\underset{\sim}{c}$ are non-parallel and $\underset{\sim}{b}$ is non-zero, $\underset{\sim}{a}+\underset{\sim}{c}$ // $\underset{\sim}{b}$ $\overrightarrow{OM}=\frac{\underset{\sim}{a}+\underset{\sim}{c}}{2}$ // $\overrightarrow{OB}$ Since O is a common point and $\overrightarrow{OM}$ // $\overrightarrow{OB}$, O, B and M are collinear. **Method 2:** $\overrightarrow{OM}\times\overrightarrow{OB}=\left(\frac{\underset{\sim}{a}+\underset{\sim}{c}}{2}\right)\times\underset{\sim}{b}$ $=\frac{1}{2}(\underset{\sim}{a}\times\underset{\sim}{b}+\underset{\sim}{c}\times\underset{\sim}{b})$ $=\frac{1}{2}(\underset{\sim}{a}\times\underset{\sim}{b}-\underset{\sim}{b}\times\underset{\sim}{c})$ $=\underset{\sim}{0}$ $\quad(\text{given } \underset{\sim}{a}\times\underset{\sim}{b}=\underset{\sim}{b}\times\underset{\sim}{c})$ Hence $\overrightarrow{OM}$ // $\overrightarrow{OB}$. Since $\overrightarrow{OM}$ // $\overrightarrow{OB}$ and O is a common point, O, B and M are collinear.

(b)

Given angle between $\underset{\sim}{p}$ and $\underset{\sim}{q}$ is $\cos^{-1}\left(\frac{1}{6}\right)$

$$\therefore \underset{\sim}{p}\cdot\underset{\sim}{q} = \left|\underset{\sim}{p}\right|\left|\underset{\sim}{q}\right|\cos\left[\cos^{-1}\left(\frac{1}{6}\right)\right]$$

$$\underset{\sim}{p}\cdot\underset{\sim}{q} = \frac{1}{6}\left|\underset{\sim}{p}\right| \qquad \left(\text{given } \underset{\sim}{q} \text{ is a unit vector}\right)$$

$$\left|\underset{\sim}{p}+10\underset{\sim}{q}\right| = 2\left|\underset{\sim}{p}-5\underset{\sim}{q}\right|$$

$$\left|\underset{\sim}{p}+10\underset{\sim}{q}\right|^2 = 4\left|\underset{\sim}{p}-5\underset{\sim}{q}\right|^2$$

$$\left(\underset{\sim}{p}+10\underset{\sim}{q}\right)\cdot\left(\underset{\sim}{p}+10\underset{\sim}{q}\right) = 4\left(\underset{\sim}{p}-5\underset{\sim}{q}\right)\cdot\left(\underset{\sim}{p}-5\underset{\sim}{q}\right)$$

$$\left|\underset{\sim}{p}\right|^2 + 20\underset{\sim}{p}\cdot\underset{\sim}{q} + 100\left|\underset{\sim}{q}\right|^2 = 4\left|\underset{\sim}{p}\right|^2 - 40\underset{\sim}{p}\cdot\underset{\sim}{q} + 100\left|\underset{\sim}{q}\right|^2$$

$$3\left|\underset{\sim}{p}\right|^2 - 60\underset{\sim}{p}\cdot\underset{\sim}{q} = 0$$

$$3\left|\underset{\sim}{p}\right|^2 - 60\left(\frac{1}{6}\left|\underset{\sim}{p}\right|\right) = 0$$

$$3\left|\underset{\sim}{p}\right|^2 - 10\left|\underset{\sim}{p}\right| = 0$$

$$\left|\underset{\sim}{p}\right| = \frac{10}{3} \quad \left(\because \left|\underset{\sim}{p}\right| > 0\right)$$

4	**Solution**
(a) (i)	$\ln y + xy = \cos x \quad ---(1)$ Differentiate w.r.t. x, $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} + y + x\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin x \quad ---(2)$ Differentiate w.r.t. x, $\frac{1}{y}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{1}{y^2}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\mathrm{d}y}{\mathrm{d}x} + x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\cos x \quad ---(3)$ When $x = 0$, From (1): $\ln y = 1 \Rightarrow y = \mathrm{e}$ From (2): $\frac{1}{\mathrm{e}}\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{e} = 0 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\mathrm{e}^2$ From (3): $\frac{1}{\mathrm{e}}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{1}{\mathrm{e}^2}\left(-\mathrm{e}^2\right)^2 - 2\mathrm{e}^2 = -1$ $\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \mathrm{e}\left(3\mathrm{e}^2 - 1\right)$ $y = \mathrm{e} - \mathrm{e}^2x + \frac{\mathrm{e}\left(3\mathrm{e}^2-1\right)}{2}x^2 + \ldots$

ii)	To find the first term of the series, we need to substitute $x = 0$, however $\operatorname{cosec} 0$ is undefined. Hence, no Maclaurin's series for y exist.
(b)	$\dfrac{\cos\left(x-\dfrac{3\pi}{4}\right)}{\sqrt{2-2x}}$ $=\dfrac{\cos x\cos\dfrac{3\pi}{4}+\sin x\sin\dfrac{3\pi}{4}}{\sqrt{2-2x}}$ $=\dfrac{1}{\sqrt{2}}(-\cos x+\sin x)(2-2x)^{-\frac{1}{2}}$ $\approx\dfrac{1}{\sqrt{2}}\left(-\left(1-\dfrac{x^2}{2}\right)+x\right)(2)^{-\frac{1}{2}}(1-x)^{-\frac{1}{2}}$ (using small angle approx) $=\dfrac{1}{2}\left(-1+x+\dfrac{x^2}{2}\right)\left(1+\left(-\dfrac{1}{2}\right)(-x)+\dfrac{\left(-\dfrac{1}{2}\right)\left(-\dfrac{3}{2}\right)}{2!}(-x)^2+\ldots\right)$ $=\dfrac{1}{2}\left(-1+x+\dfrac{x^2}{2}\right)\left(1+\dfrac{x}{2}+\dfrac{3x^2}{8}+\ldots\right)$ $=\dfrac{1}{2}\left(-1-\dfrac{x}{2}-\dfrac{3x^2}{8}+x+\dfrac{x^2}{2}+\dfrac{x^2}{2}+\ldots\right)$ $=\dfrac{1}{2}\left(-1+\dfrac{x}{2}+\dfrac{5x^2}{8}+\ldots\right)$ $\approx\dfrac{1}{16}(5x^2+4x-8)$

5	**Solution**
(a)	$y=\dfrac{-2x^2-5x+a}{x-2}$ $y(x-2)=-2x^2-5x+a$ $2x^2+(y+5)x+(-2y-a)=0$ For x to not have real roots, discriminant < 0 $(y+5)^2-4(2)(-2y-a)<0$ $y^2+10y+25+16y+8a<0$ $y^2+26y+25+8a<0$ $(y+13)^2-169+25+8a<0$ $(y+13)^2-(144-8a)<0$ $\left(y+13+\sqrt{144-8a}\right)\left(y+13-\sqrt{144-8a}\right)<0$ $\left[y-\left(-13-\sqrt{144-8a}\right)\right]\left[y-\left(-13+\sqrt{144-8a}\right)\right]<0$ $-13-\sqrt{144-8a}<y<-13+\sqrt{144-8a}$ The two values are: $-13-2\sqrt{36-2a}$ and $-13+2\sqrt{36-2a}$

(b)

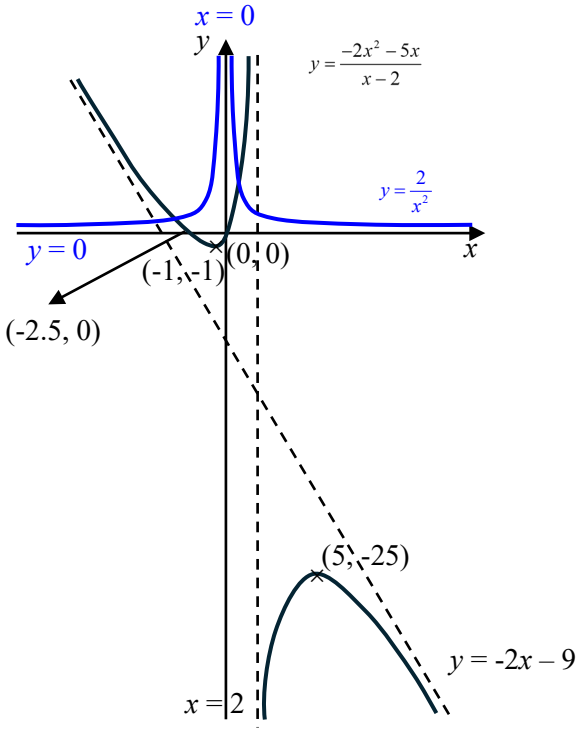
$y = \dfrac{-2x^2 - 5x}{x - 2} = -2x - 9 - \dfrac{18}{x - 2}$

Equations of asymptotes are $y = -2x - 9$ and $x = 2$

(c)

Turning points (by GC): (-1, -1) and (5, -25)

Axial intercepts (by GC): (0, 0) and (-2.5, 0)



(d)

$2x^4 + 5x^3 + 2x - 4 = 0$

$2x - 4 = -2x^4 - 5x^3$

$2(x - 2) = x^2(-2x^2 - 5x)$

$\dfrac{2}{x^2} = \dfrac{-2x^2 - 5x}{x - 2}$

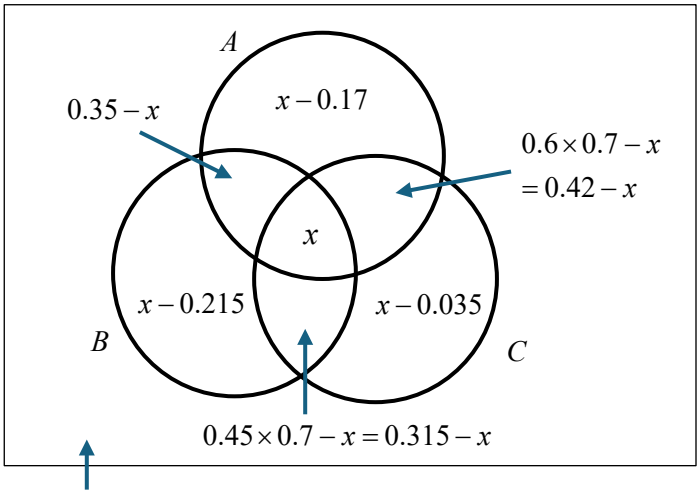
Add $y = \dfrac{2}{x^2}$ into part **(c)**. From the graph, the equation has 2 real roots.

6	**Solution**
(a)	Let X and Y be the number of defective microchips in the first and second sample of n respectively. $X \sim \mathrm{B}(n,\ 0.2)$ and $Y \sim \mathrm{B}(n,\ 0.2)$ $X > 2 \Rightarrow$ Batch rejected $X = 2 \Rightarrow \begin{cases} Y \geq 1 \Rightarrow \text{Batch rejected} \\ Y = 0 \Rightarrow \text{Batch accepted} \end{cases}$ $X \leq 1 \Rightarrow$ Batch accepted

	$P = \mathrm{P}(X \le 1) + \mathrm{P}(X = 2)\mathrm{P}(Y = 0)$ $= \mathrm{P}(X = 0) + \mathrm{P}(X = 1) + \mathrm{P}(X = 2)\mathrm{P}(Y = 0)$ $= (0.8)^n + \binom{n}{1}(0.2)(0.8)^{n-1} + \binom{n}{2}(0.2)^2(0.8)^{n-2}(0.8)^n$ $= [0.8 + 0.2n](0.8)^{n-1} + \frac{n(n-1)}{2}(0.04)(0.8)^{2n-2}$ $= (0.8 + 0.2n)(0.8)^{n-1} + 0.02n(n-1)(0.8)^{2n-2}$
(b)	$(0.8 + 0.2n)(0.8)^{n-1} + 0.02n(n-1)(0.8)^{2n-2} < 0.1$ From GC, n | P 18 | 0.102 19 | 0.0851 Least $n = 19$
(c)	Let W be the number of batches (out of 100) accepted $W \sim \mathrm{B}(100,\ 0.398)$ $\mathrm{P}(40 \le W \le 45) = \mathrm{P}(W \le 45) - \mathrm{P}(W \le 39)$ $= 0.399$ (3 s.f)

7	**Solution**
(a)	$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)}$ $\mathrm{P}(A) = \frac{\frac{7}{20}}{\frac{7}{12}} = \frac{3}{5}$ $\mathrm{P}(A \cap B') = \mathrm{P}(A) - \mathrm{P}(A \cap B) = \frac{3}{5} - \frac{7}{20} = \frac{1}{4}$
(b)	$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \quad \Rightarrow \quad \mathrm{P}(B) = \frac{\frac{7}{20}}{\frac{7}{9}} = \frac{9}{20}$ $\mathrm{P}(A \cup B) = \mathrm{P}(B) + \mathrm{P}(A \cap B') = \frac{9}{20} + \frac{1}{4} = \frac{7}{10}$ **OR** $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B) = \frac{3}{5} + \frac{9}{20} - \frac{7}{20} = \frac{7}{10}$
(c)	$\mathrm{P}(A' \cap B') = 1 - \mathrm{P}(A \cup B) = \frac{3}{10}$

(d)



$\mathrm{P}(A'\cap B'\cap C')=1-\mathrm{P}(A\cup B\cup C)=0.335-x$

$x-0.215\geq 0 \quad\Rightarrow\quad x\geq 0.215$

$0.315-x\geq 0 \quad\Rightarrow\quad x\leq 0.315$

Greatest and least possible values of $\mathrm{P}(A\cap B\cap C)$ are 0.315 and 0.215 respectively.

8	**Solution**

(a)

Method 1

$$\mathrm{P}(X=3)=\mathrm{P}(R,R',R)\times 2!+\mathrm{P}(W,W',W)\times 2!+\mathrm{P}(B,B',B)\times 2!$$

$$=\left(\frac{6}{14}\right)\left(\frac{5+3}{13}\right)\left(\frac{5}{12}\right)\times 2!+\left(\frac{5}{14}\right)\left(\frac{6+3}{13}\right)\left(\frac{4}{12}\right)\times 2!+\left(\frac{3}{14}\right)\left(\frac{6+5}{13}\right)\left(\frac{2}{12}\right)\times 2!$$

$$=\frac{972}{2184}$$

$$=\frac{81}{182}\quad\text{(Shown)}$$

Method 2

$$\mathrm{P}(X=3)=\mathrm{P}(RW_)\times 2!+\mathrm{P}(BW_)\times 2!+\mathrm{P}(RB_)\times 2!$$

$$=\left(\frac{6}{14}\right)\left(\frac{5}{13}\right)\left(\frac{5+4}{12}\right)\times 2!+\left(\frac{3}{14}\right)\left(\frac{5}{13}\right)\left(\frac{2+4}{12}\right)\times 2!+\left(\frac{6}{14}\right)\left(\frac{3}{13}\right)\left(\frac{5+2}{12}\right)\times 2!$$

$$=\frac{972}{2184}$$

$$=\frac{81}{182}\quad\text{(Shown)}$$

$$\mathrm{P}(X=2)=\mathrm{P}(RR)+\mathrm{P}(WW)+\mathrm{P}(BB)$$

$$=\left(\frac{6}{14}\right)\left(\frac{5}{13}\right)+\left(\frac{5}{14}\right)\left(\frac{4}{13}\right)+\left(\frac{3}{14}\right)\left(\frac{2}{13}\right)\quad\text{OR}\quad \frac{{}^{6}C_{2}}{{}^{14}C_{2}}+\frac{{}^{5}C_{2}}{{}^{14}C_{2}}+\frac{{}^{3}C_{2}}{{}^{14}C_{2}}$$

$$=\frac{4}{13}$$

$$\mathrm{P}(X=4)=1-\frac{81}{182}-\frac{4}{13}=\frac{45}{182}$$

OR

$$\mathrm{P}(X=4)=\mathrm{P}(RWB_)=\left(\frac{6}{14}\right)\left(\frac{5}{13}\right)\left(\frac{3}{12}\right)\times 3!=\frac{45}{182}$$

x	2	3	4
$\mathrm{P}(X=x)$	$\frac{4}{13}$	$\frac{81}{182}$	$\frac{45}{182}$

(b)

$\mathrm{E}(X)=2.9396$

$\mathrm{Var}(X)=0.55129$

$\mathrm{E}(200-50X)=200-50\mathrm{E}(X)=53.02$

$\mathrm{Var}(200-50X)=2500\mathrm{Var}(X)=1378.23$

Expectation and variance of the prize money won by the player at the end of the game are \$53.02 and $\$^2$1378.23

OR

Let Y be the winning in a game

Y	100	50	0
$\mathrm{P}(Y=y)$	$\frac{4}{13}$	$\frac{81}{182}$	$\frac{45}{182}$

$\mathrm{E}(Y)=53.02$

$\mathrm{Var}(Y)=1378.23$

Expectation and variance of the prize money won by the player at the end of the game are \$53.02 and $\$^2$1378.23

(c)

Expected total prize money is $\$\left[53.0220+\frac{4}{13}(53.0220)\right]=\$69.34$

OR

Let W be the winning in new game

W	100 + 53.02	50	0
$\mathrm{P}(W=w)$	$\frac{4}{13}$	$\frac{81}{182}$	$\frac{45}{182}$

$\frac{4}{13}(100+53.0220)+50\left(\frac{81}{182}\right)=69.34$

Expected total prize money is \$69.34

(d)

9! ways to arrange 9 items: 1 pair + 3 players + 5 empty seats

2! ways to permutate the pair

Method 1: No. of arrangements $=\frac{9!2!}{5!}=6048$

5 identical empty seats

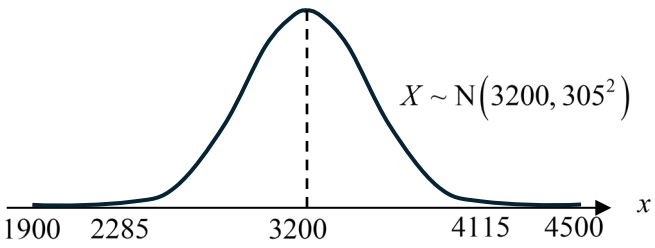
9 possible adjacent pairs of seats

Choose 3 out of 8 remaining seats for the 3 players

Method 2: No. of arrangements $= 9 \times (2!) \times {}^{8}C_{3} \times 3! = 6048$

2! ways to permutate the pair

3! ways to permutate the 3 players

9	**Solution**
(a)	Let X mAh be the capacity of a Standard battery cell. $X \sim \mathrm{N}(\mu, \sigma^2)$ $\mathrm{P}(X < 2750) = \mathrm{P}(X > 3650)$ By symmetry, $\mu = \dfrac{2750 + 3650}{2} = 3200$
(b)	$\mathrm{P}(2750 < X < 3650) = 0.860$ $\mathrm{P}\left(\dfrac{2750 - 3200}{\sigma} < Z < \dfrac{3650 - 3200}{\sigma}\right) = 0.860$ From the GC, $\mathrm{P}(-1.4758 < Z < 1.4758) = 0.860$ $\dfrac{2750 - 3200}{\sigma} = -1.4758$ $\sigma = 304.92 \approx 305 \quad (\text{to 3 s.f.})$
(c)	
(d)	Let Y mAh be the capacity of a Premium battery cell $Y \sim \mathrm{N}(3520, 112560.25)$ $\mathrm{E}(Y) = \mathrm{E}(1.1X) = 1.1(3200) = 3520$ $\mathrm{Var}(Y) = \mathrm{Var}(1.1X) = 1.1^2(305^2) = 112560.25$ Let $T = X_1 + X_2 - Y$ $\mathrm{E}(T) = \mathrm{E}(X_1 + X_2 - Y) = 2(3200) - 3520 = 2880$ $\mathrm{Var}(T) = \mathrm{Var}(X_1 + X_2 - Y) = 2(305^2) + 112560.25$ $= 298610.25$ $T \sim \mathrm{N}(2880, 298610.25)$ $\mathrm{P}(T > 3000) = 0.413 \,(\text{to 3 s.f.})$

(e)	Let B mAh be the capacity of a Budget battery cell $\mathrm{E}(B)=1200;\quad \mathrm{Var}(B)=400^2$ Since $n=100$ is large, by Central Limit Theorem, $\overline{B}\sim \mathrm{N}\left(1200, \frac{400^2}{100}\right)$ approximately $\mathrm{P}\left(\overline{B}<1150\right)=0.10565$ Let W be the number of samples have mean capacity of less than 1150 mAh. $W\sim \mathrm{B}(3, 0.10565)$ Required probability = $\mathrm{P}(W=2)=0.0299$ **OR** Required probability $=\binom{3}{2}\left[\mathrm{P}\left(\overline{B}<1150\right)\right]^2 \mathrm{P}\left(\overline{B}>1150\right)$ $=0.0299$ (to 3 s.f.)

10	**Solution**
(a)	A random sample in this context means that each bottle of tomato juice produced has an equal chance of being selected, and the selections are independent of each other.
(bi)	Let X ml be the volume of tomato juice in a bottle and μ be the population mean of X. $\mathrm{H}_0: \mu=250$ $\mathrm{H}_1: \mu<250$ Perform a 1 tail test at 5% level of significance **Assumption:** We assume X (i.e. volume of tomato juice in a bottle) follows a normal distribution. Under H_0, $\overline{X}\sim \mathrm{N}\left(250, \frac{5^2}{12}\right)$ From the sample, $\bar{x}=247.65$ (exact), p-value $=\mathrm{P}(\overline{X}\le 247.65)=0.0517$ (to 3 s.f.) Since p-value $=0.0517>0.05$, we do not reject H_0 and conclude that there is insufficient evidence at the 5% significance level that the population mean volume of tomato juice in a bottle is less than 250 ml or that it has been overstated.
(bii)	$p-$value = 0.0517 means that 5.17% is the smallest level of significance at **which the test concludes** that the **population mean** volume of tomato juice in a bottle is less than 250 ml using this sample.
(c)	Let Y ml be the volume of prune juice in a bottle and μ ml be the population mean of Y. $\mathrm{H}_0: \mu=k$ $\mathrm{H}_1: \mu\neq k$

Perform a 2-tail test at 2% level of significance

$n = 60, \quad \bar{y} = 179.3, \quad s^2 = \frac{60}{59}(50.8)$

Since n is large, by Central Limit Theorem $\overline{Y} \sim \mathrm{N}\left(k, \frac{\frac{60(50.8)}{59}}{(60)}\right)$ approximately.

Since H_0 is rejected, p-value ≤ 0.02

$\mathrm{P}\left(\overline{Y} \leq 179.3\right) \leq 0.01$ or $\mathrm{P}\left(\overline{Y} \geq 179.3\right) \leq 0.01$

$\mathrm{P}\left(Z \leq \frac{179.3 - k}{\sqrt{\frac{50.8}{59}}}\right) \leq 0.01$ or $\mathrm{P}\left(Z \geq \frac{179.3 - k}{\sqrt{\frac{50.8}{59}}}\right) \leq 0.01$

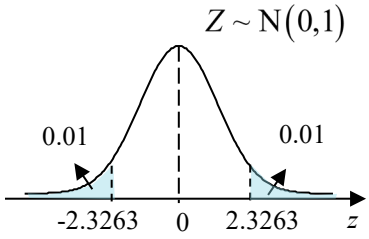
From GC, $\mathrm{P}\left(Z \leq -2.3263\right) = 0.01$ or $\mathrm{P}\left(Z \geq 2.3263\right) = 0.01$

From the diagram,

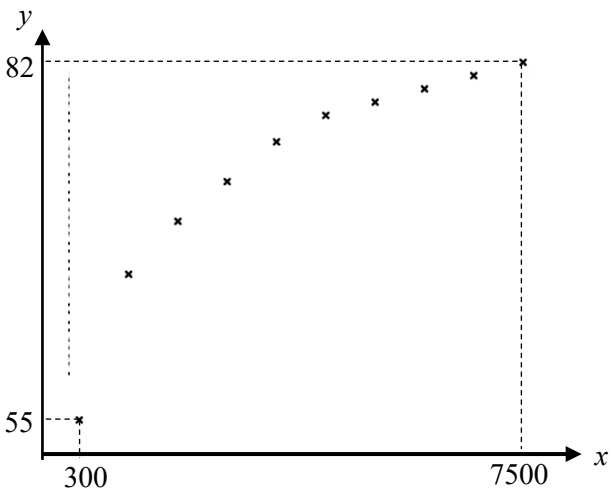
$\frac{179.3 - k}{\sqrt{\frac{50.8}{59}}} \leq -2.3263$ or $\frac{179.3 - k}{\sqrt{\frac{50.8}{59}}} \geq 2.3263$

$k \geq 181.46$ or $k \leq 177.14$

$\therefore \left\{k \in \mathbb{R}^+ : k \leq 177 \quad \text{or} \quad k \geq 181\right\}$



11	**Solution**
(a)	$\bar{x} = 3900$ $\bar{y} = 0.0031667\bar{x} + 61.65$ $= 0.0031667(3900) + 61.65$ $= 74.00013$ (exact) $\frac{55 + 66 + 70 + 73 + 76 + 79 + 80 + 81 + 82 + k}{10} = 74.00013$ $622 + k = 740.0013$ $k = 78.0013$ ≈ 78

(b)	
(c)	From the scatter diagram, as x increases, y increases at a decreasing rate. Therefore, relationship between x and y should not be modelled by the equation $y = ax + b$
(d)	From the GC, $\ln y = 3.3126 + 0.12369 \ln x$ $\ln y = 3.31 + 0.124 \ln x \quad (\text{to 3 s.f.})$ $r = 0.998 \quad (\text{to 3 s.f.})$
(e)	Substitute $y = \dfrac{6000}{100} = 60$: $\ln(60) = 3.3126 + 0.12369 \ln x$ $\ln x = 6.3202$ $x = 555.68$ Estimated advertising budget $= 555.68 \times 100 \approx \$55600 \quad (\text{nearest hundred})$ Since $y = 60$ lies within the range $55 \le y \le 82$ and $r = 0.998$ is close to 1 which means the model is a good fit, the estimate is reliable.
(e)	Dilip's conclusion is invalid. Correlation does not imply causation. There could be **other factors driving both variables**, such as a general improvement in economic conditions or seasonal trends.